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Please write clearly in block capitals.

Centre number | | | | | | Candidate number | | | | |

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Functional Skills Certificate
FUNCTIONAL MATHEMATICS

Level 2

Thursday 8 November 2018 Morning Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:
- a calculator
- mathematical instruments
- a copy of the Data Book (Examination) (enclosed).



For Examiner's Use	
Question	Mark
1	
2	
3	
4	
TOTAL	

Instructions
- Use black ink or black ball-point pen. Draw diagrams in pencil.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
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· Assessment: November 2018

· Subject: AQA Functional Skills Maths Level 2

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Answer **all** questions in the spaces provided.

outside the box

1 **Angling**

There is a **data sheet** for Angling.

1 (a) In an angling competition there are seven quarter-finals.

The table shows the number of anglers in each of the quarter-finals.

Quarter-final 1	15 anglers	Semi-final 1	Final
Quarter-final 2	12 anglers		
Quarter-final 3	12 anglers		
Quarter-final 4	9 anglers		
Quarter-final 5	12 anglers	Semi-final 2	
Quarter-final 6	12 anglers		
Quarter-final 7	12 anglers		

How many anglers qualify for the final?

You **must** show your working.

[4 marks]

QF1: 5, QF2: 4, QF3: 4, QF4: 3, QF5: 4, QF6: 4, QF7: 4.

⇒ SF1 (in comp) = 5 + 4 + 4 + 3 = 16 ⇒ 4 qualify for Final

SF2 (in comp) = 4 + 4 + 4 = 12 ⇒ 3 qualify for Final

⇒ 4 + 3 = 7 Finalists.

1 (b) John and Lata are in one of the quarter-finals.
Here are the weights of the fish they have caught in the first 12 hours.

John	17 lb 2 oz	19 lb 12 oz	16 lb 13 oz	20 lb 5 oz
Lata	18 lb 4 oz	17 lb 1 oz	18 lb 11 oz	

John says,

"The mean weight of my fish is **more than** the mean weight of Lata's fish."

Is he correct?
You **must** show your working.

[5 marks]

In ounces:

John: 274, 316, 269, 325

Lata: 292, 273, 299

Total:

John: $274 + 316 + 269 + 325 = 1184$.

Lata: $292 + 273 + 299 = 864$

Average:

John: $\frac{1184}{4} = 296$oz

Lata: $\frac{864}{3} = 288$oz.

Yes, he is correct.

box

1 (c) The weight of the heaviest fish that John catches is 20 lb 3 oz
Last year, in Australia, he caught a fish weighing 9 kg 120 g

John says,

"The fish I caught in Australia was heavier."

Is he correct?
You **must** show your working.

[5 marks]

(20×16)

20 lb 3oz = 320 + 3 = 323oz

323 × 28.35 = 9157.05g

= 9kg 157.05g.

No, he is incorrect.

14

Do not write outside the box

2 Council Tax

There is a **data sheet** for Council Tax.

2 (a) Tom and Mary are both 47 years old.

They share a home in Kribly.

The Council Tax is band **G**.

They pay their Council Tax in 10 monthly payments.

Work out the Council Tax payment for April.

[3 marks]

$\frac{2977.11}{10} = £297.711 \rightarrow £298$.

£2977.11 − (9 × £298) = £295.11

Question 2 continues on the next page

box

2 (b) Dan is disabled and lives on his own.
His home is **not** adapted for his needs.
He moves to a new home.
He makes these notes.

	Old home	New home
Area	Heath	Mossett
Usual band	**C**	**D**
25% discount	Yes	Yes

Dan will adapt his new home in Mossett for his needs.

He says,

"My Council Tax will then be cheaper than I paid in Heath by at least £100"

Is he correct?
You **must** show your working.

[6 marks]

Old Home: £1495.44

£1495.44 × $\frac{25}{100}$ = £373.86.

£1495.44 − £373.86 = £1121.58

New Home: After re-banding to C, £1315.45

£1315.45 × $\frac{25}{100}$ = £328.8625

£1315.45 − £328.8625 = £986.5875

£1121.58 − £986.5875 = £134.99. Yes, he is correct.

2 (c) In Waybridge, the total income from Council Tax is £38 320 000

Estimate how much of the £38 320 000 the council spends on buses each week.
Give your answer to a suitable degree of accuracy.

[3 marks]

£38 320 000 × 0·08 = £3065600 per year

⇒ $\frac{£3065600}{52}$ = £58953.85 per week.

→ £59000 per week.

Do not write outside the box

David is calculating the Council Tax in Bretton, another part of Waybridge.

The Council Tax for band **D** is £1504.44

He works out the Council Tax in the other bands using these proportions.

	A	B	C	D	E	F	G	H
Proportion of band **D**	$\frac{6}{9}$	$\frac{7}{9}$	$\frac{8}{9}$	1	$\frac{11}{9}$	$\frac{13}{9}$	$\frac{15}{9}$	$\frac{18}{9}$

†2 **(d)** Work out the Council Tax in band **B**.

[2 marks]

£1504.44 × $\frac{7}{9}$ = £1170.12

Check your answer.

Show how you have done your check.

[1 mark]

$\frac{£1170.12 \times 9}{7}$ = £1504.44.

2 (e) How many times bigger is the Council Tax in band **H** than in band **A**?

Circle your answer.

[1 mark]

$\frac{12}{9}$ 2 (3) 12

16

Turn over for the next question

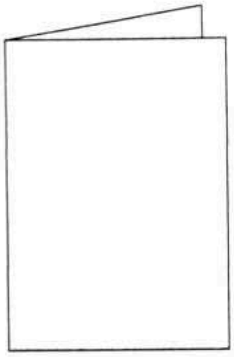
outside the box

3 **Christmas Cards**



I make and sell Christmas cards.
I put designs onto card blanks.

Leah



Card blank



Card blank with design

Do not write outside the box

†3 (a) You buy card blanks in packs.

Card Blanks
Packs of 15
Only £6.49

Work out the cost of 60 of these card blanks.

[2 marks]

$\frac{60}{15} \times £6.49 = £25.96.$

Check your answer.
Show how you have done your check.

[1 mark]

$\frac{£25.96}{£6.49} \times 15 = 60.$

Question 3 continues on the next page

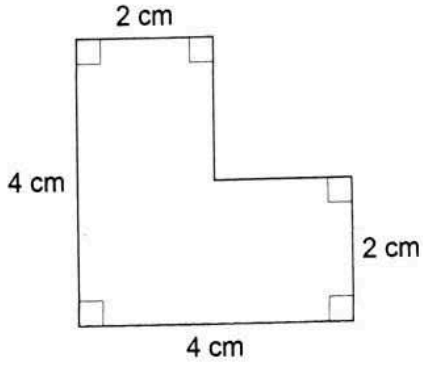
box

Leah is designing a card with a chimney and a **pair** of Santa's boots.



3 (b) Leah cuts the boots from sheets of paper measuring 13 cm by 10 cm

Here is a sketch of one of the boots.



Not drawn accurately

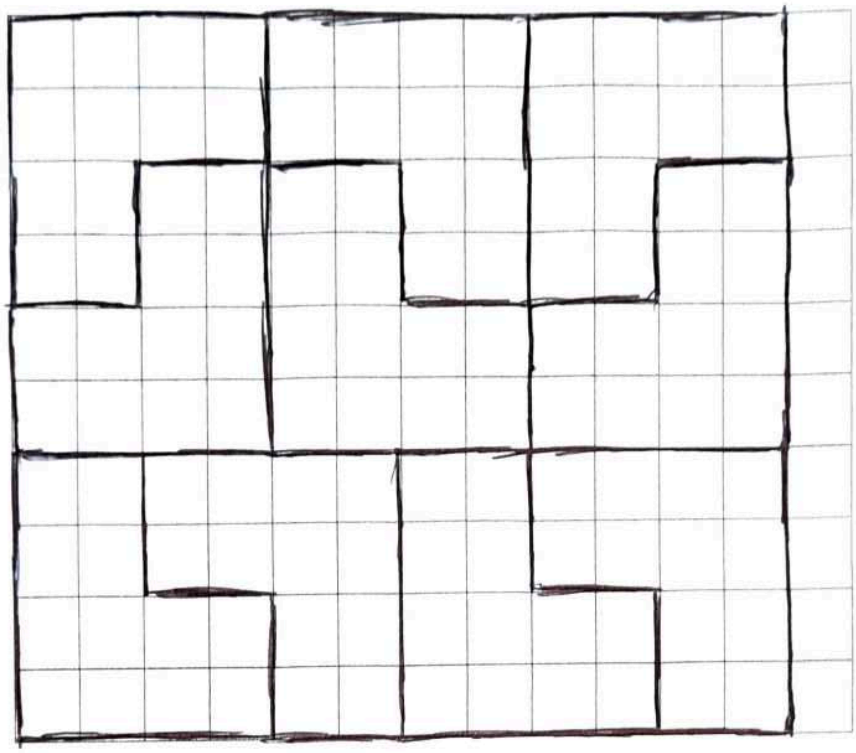
Show how 10 boots can be cut from a 13 cm by 10 cm sheet.

Use the grids opposite.

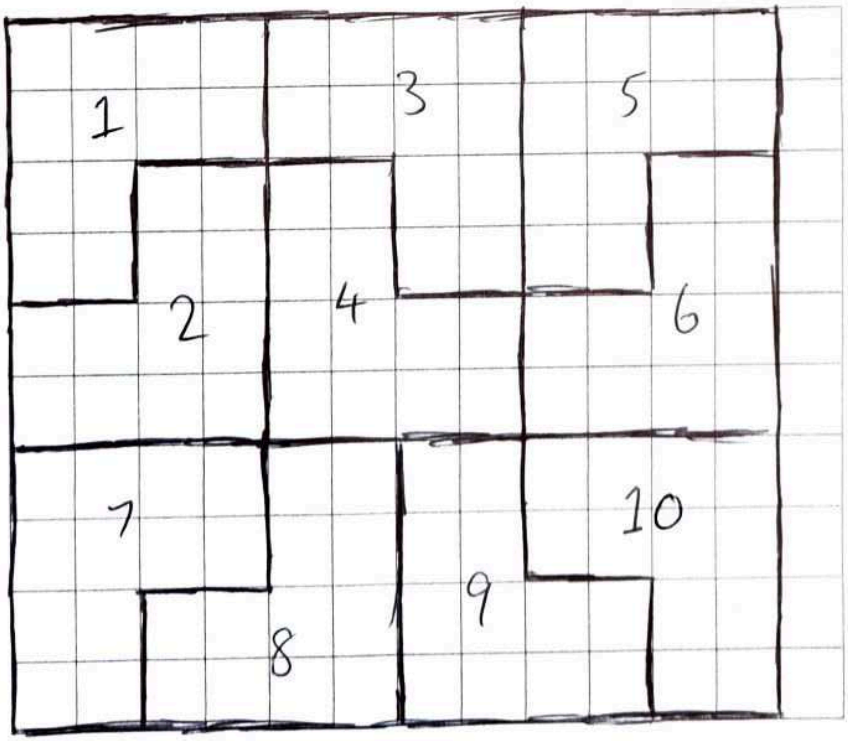
[3 marks]

outside the box

Practise on this grid.



Put your answer on this grid.



Turn over ▶

outside the box

3 (c) Leah makes 150 cards with this design.



She uses 15 sheets of red felt for the chimneys.

She buys a bulk pack of 150 card blanks.

This table shows the costs of the items she needs for the cards.

Item	Cost
Paper for boots	75p per sheet
Red felt	£1.40 per sheet
150 card blanks	£43.50
Other costs	£16.50

Leah sells each card for £1.20

She says,

"My profit will be **more than** 65% of the cost of making the cards."

Is she correct?

You **must** show your working.

[8 marks]

Boots: $\frac{150}{10} \times 2 \times 75p = 2 \times 1125p = £11.25 \times 2 = £22.50$

Red felt: $15 \times £1.40 = £21.00$.

Card blanks: £43.50, Other costs: £16.50

£22.50 + £21.00 + £43.50 + £16.50 = £103.50

box

£1.20 x £150 = £180

£180 − £103.50 = £76.50.

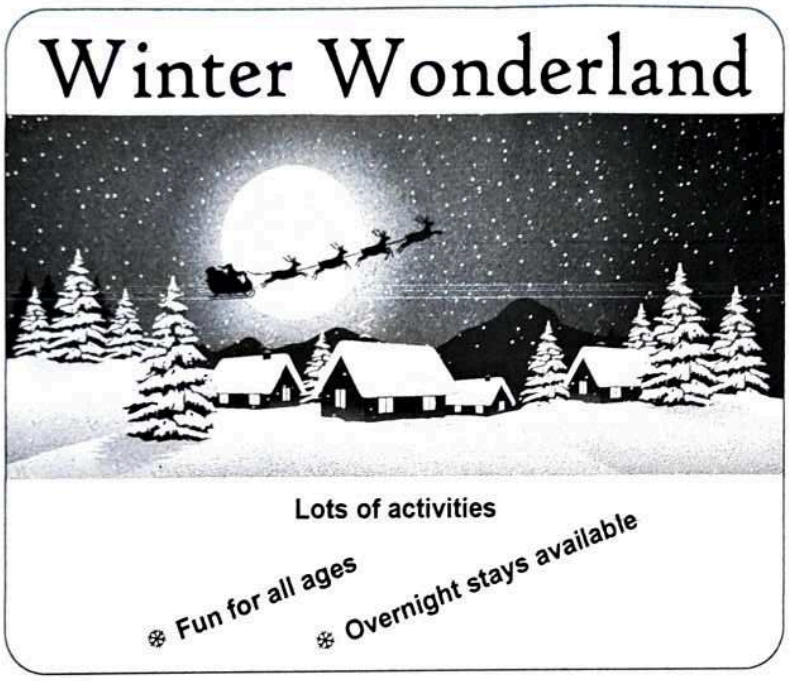
$\frac{£76.50}{£103.50}$ = 0.739 = 73.9 %.

Yes, she is correct.

14

outside the box

4 **Winter Wonderland**



This is Jane and her family.



Jane



Mike



Jack, aged 8



Meg, aged 4

Do not write outside the box

Jane and her family are planning a visit to Winter Wonderland.

4 (a) Here are the ticket prices.

Ticket Prices

£38 per adult or child aged 5 and over

£25 per child aged under 5

plus £5 booking fee per family

Altogether, how much will Jane pay for herself and her family?

[3 marks]

£38 × 3 = £114.

£114 + £25 + £5 = £144.

4 (b) The activities at Winter Wonderland start at 10 am

Jane and her family want to arrive 45 minutes before the first activity starts.

They live 80 miles away.

Jane will drive at an average speed of 50 miles per hour.

What is the **latest** time they should set off?

[4 marks]

10:00am − 45mins → 9:15am arrival.

80mi

50mph = 1.6hrs. = 96mins

9:15am − 96mins → 7:39am.

4 (c) Jane has already booked the following activities.

Do not write outside the box

Time	Activity	Length of activity
10 00	Christmas theatre	40 minutes
13 30	Mrs Christmas	40 minutes
16 10	Elves	30 minutes

She has also booked lunch at the Boat Inn from 12 30 to 13 20

Jane is planning other activities for the rest of the day.



We need at least 10 minutes between activities and before and after lunch.

We want to do as many different activities as possible.

We will leave after the Elves activity.

Here are the other activities.

Each activity can start at any time.

Activity	Length
Snow tubing	40 minutes
Reindeer	30 minutes
Ice skating	50 minutes
Sleigh ride	20 minutes
Winter crafts	30 minutes
Treasure hunt	40 minutes

outside the box

Write a plan for the day, showing the start and end times of each activity.

[5 marks]

10:00am – 10:40am – Xmas Theatre

10:50am – 11:30am – Snow Tubing

11:40 am – 12:20 pm – Treasure Hunt

12:30pm – 13:20pm – Lunch

13:30 pm – 14:10pm – Mrs Christmas

14:20 pm – 14:40 – Sleigh Ride

14:50 – 15:20 – Winter Crafts

15:30 – 16:00 – Reindeer

16:10 – 16:40 – Elves

Do not write outside the box

4 (d) Myra and some friends are going to Winter Wonderland.
They will stay for 3 nights.

Myra looks at the prices of different-sized lodges.

Arrival date	Size of Lodge		
	Sleeps 4	Sleeps 5	Sleeps 6
Friday 23rd Nov (3 nights)	£320	£385	£455
Friday 30th Nov (3 nights)	£378	£439	£495
Friday 7th Dec (3 nights)	£410	£450	£545

Myra is booking lodges for 20 people.
They will arrive on Friday 30th November.

What is the cheapest possible price for her booking?
State clearly how many of each type of lodge to book.

[4 marks]

Getting a 4-bed and a 6-bed lodge is cheaper than two 5-bed lodges.

2 x 4-bed + 2 x 6-bed : (2 x 378) + (2 x 495)
= 756 + 990
= £1746

5 x 4-bed : 5 x 378 = £1890.

The cheapest option is two 4-bed and two 6-bed lodges, coming to £1746.

16

END OF QUESTIONS

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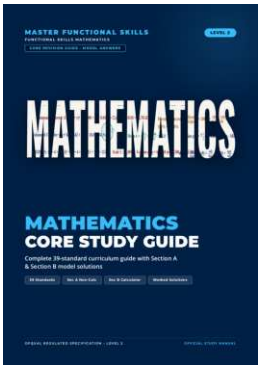
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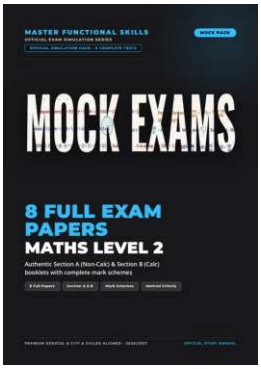


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